

Prolog Program for Fruit and its Color Using Back Tracking

1. Aim

To write and execute a Prolog program that stores fruits and their colors, and uses backtracking to find all fruits matching a given color.

2. Objective

- To understand how Prolog's backtracking explores multiple solutions to a query.
- To represent fruit-color associations as facts.
- To retrieve all fruits of a given color by forcing backtracking over alternatives.
- To use findall/3 to collect every backtracked solution into a single list.

3. Algorithm

1. Start the program.
2. Create a Prolog database using the predicate fruit/2, storing each fruit with its color.
3. Define a rule find_by_color/2 that matches a fruit against a given color.
4. Enter the color as a query with an unbound Fruit variable.
5. Prolog attempts to unify with the first matching fact.
6. On backtracking (requesting more solutions), Prolog retries with the next matching fact.
7. Use findall/3 to collect all fruits of the given color into a single list in one query.
8. Display the matching fruit(s).
9. Stop the program.

4. Flowchart



5. Code with Output

Code:

```
% -----  
% Fruit - Color Database  
% fruit(Name, Color)  
% -----  
fruit(apple, red).  
fruit(banana, yellow).  
fruit(grape, purple).  
fruit(cherry, red).  
fruit(mango, yellow).  
  
% -----  
% Rule: find a fruit of a given color  
% (backtracks through all matching facts)  
% -----  
find_by_color(Color, Fruit) :-  
    fruit(Fruit, Color).  
  
% Collect all matches at once  
all_by_color(Color, Fruits) :-  
    findall(Fruit, fruit(Fruit, Color), Fruits).
```

Output:

```
?- find_by_color(red, F).  
F = apple ;  
F = cherry ;  
false.  
  
?- all_by_color(yellow, Fruits).  
Fruits = [banana, mango].
```

6. Result

The Prolog program was successfully implemented to find fruits matching a given color, demonstrating backtracking through repeated ';' requests and bulk collection using findall/3.

7. Conclusion

Thus, the fruit-color program was successfully implemented using Prolog, illustrating how backtracking explores every clause that can satisfy a goal, one at a time, and how findall/3 gathers all such solutions in a single pass.